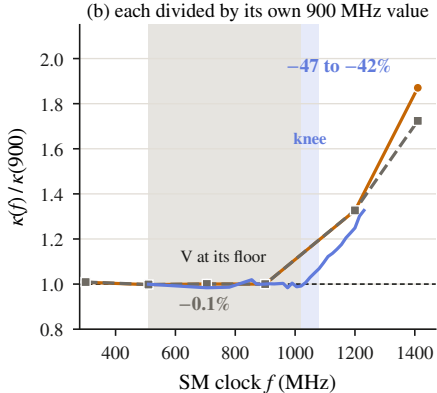
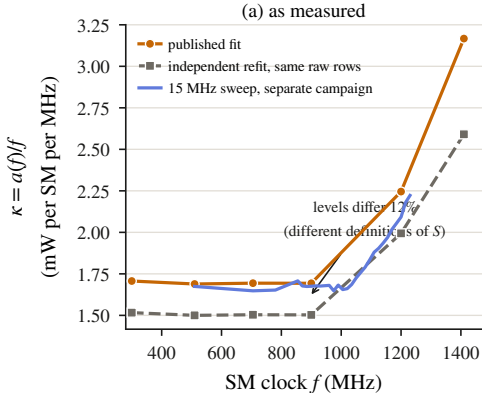




$\kappa(f) \equiv a(f)/f$, where $a(f) = \partial P_{\text{dyn}} / \partial S$ is the per-busy-SM dynamic power slope in W per SM, obtained by regressing measured dynamic

power on the number of busy SMs at fixed f . One SM draws $P_{\text{dyn}} = CV^2f$, so $\kappa = CV^2$: C is fixed by the silicon, hence κ IS a measurement of V^2 , and equally of the energy per unit of work, since W operations cost $E = P_{\text{dyn}}T = (CV^2f)(W/f) = CV^2W$ with f cancelling.

WHY (b) DIVIDES BY 900 MHz: the three curves estimate the same quantity but count S differently -- the published fit uses the wave model's S_{eff} , the refit the measured probe SM count, the sweep the whole GPU -- so they sit at slightly different levels, as (a) shows. The claim is about SHAPE (flat, then rising), which a constant factor cannot change, so (b) removes it. 900 MHz is the reference because it is the highest clock still clearly on the voltage floor; any point in the flat band gives the same picture.

Sources: 1080-row squat-kernel sweep (12 GEMM shapes $\times$ 5 SM counts $\times$ 3 reps $\times$ 6 clocks), fitted as $P_{\text{dyn}} = a(f)S + b(f, \text{shape})$; and a 29-clock 15 MHz sweep.

Knee band 1020-1080: κ is within 1% of its floor to 1035 and first exceeds +5% at 1080. The 1410 point is the weakest (118 surviving rows vs 180, R^2 0.954), hence a range.